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I. Multiple Choice Questions

1. The linear momentum and position vector of the planet is perpendicular to each other at

(a) perihelion and aphelion

(b) at all points

(c) only at perihelion

(d) no point

2. If the masses of the Earth and Sun suddenly double, the gravitational force between them will

(a) remain the same

(b) increase 2 times

(c) increase 4 times

(d) decrease 2 times

3. A planet moving along an elliptical orbit is closest to the Sun at distance r_1 and farthest away at a distance of r_2. If v_1 and v_2 are linear speeds at these points respectively. Then the ratio $\frac{v_1}{v_2}$ is

(NEET 2016)

(a) $\frac{r_2}{r_1}$ (b) $\left(\frac{r_2}{r_1}\right)^2$

(c) $\frac{r_1}{r_2}$ (d) $\left(\frac{r_1}{r_2}\right)^2$

4. The time period of a satellite orbiting Earth in a cirular orbit is independent of .

(a) Radius of the orbit

(b) The mass of the satellite

(c) Both the mass and radius of the orbit

(d) Neither the mass nor the radius of its orbit

5. If the distance between the Earth and Sun were to be doubled from its present value, the number of days in a year would be

(a) 64.5 (b) 1032

(c) 182.5 (d) 730

6. According to Kepler's second law, the radial vector to a planet from the Sun sweeps out equal areas in equal intervals of time. This law is a consequence of

(a) conservation of linear momentum

(b) conservation of angular momentum

(c) conservation of energy

(d) conservation of kinetic energy

7. The gravitational potential energy of the Moon with respect to Earth is

(a) always positive

(b) always negative

(c) can be positive or negative

(d) always zero

8. The kinetic energies of a planet in an elliptical orbit about the Sun, at positions A, B and C are K_A, K_B and K_C respectively. AC is the major axis and SB is perpendicular to AC at the position of the Sun S as shown in the figure. Then

(NEET 2018)

(a) $K_A > K_B > K_C$

(b) $K_B < K_A < K_C$

(c) $K_A < K_B < K_C$

(d) $K_B > K_A > K_C$

